Case Study Report: Computer Repair Appointment and Service Management System
Designing an Institutional Information Management (IM) Framework and System Plan

Document Overview
This document serves as the formal **Information Management (IM) Framework and System Plan** for the **Computer Repair Appointment and Service Management System**. It outlines the structural design, data governance policies, security measures, and end-to-end operational workflows governing the system.

System Operational & Project Flow
Below is the complete end-to-end project workflow detailing how information moves through each phase of the repair service lifecycle:

1. **Customer Registration:**
    - Staff input client details (Name, Contact Number, Email, Address) into the database.
    - The system validates contact formats to eliminate duplicate client entries.
2. **Device Intake:**
    - Staff register the customer's device details (Device Type, Brand, Model, Serial Number, and Reported Problem) linked directly to the customer_id.
3. **Appointment Scheduling / Intake Queue:**
    - An appointment is scheduled or logged for immediate intake.
    - The system checks technician availability and assigns the request based on specialized skills.

4. **Diagnosis & Service Allocation:**

    - The assigned technician inspects the device, records diagnostic notes, selects the necessary service from the Repair_Services catalog, and establishes an estimated repair cost.

    - Status updates to **Under Diagnosis** or **Waiting for Parts** as applicable.

5. **Repair Execution:**

    - The technician completes the required hardware maintenance, OS reinstallation, or component replacement.

    - System status updates to **Under Repair** and finally **Ready for Pickup** upon completion.

6. **Billing & Payment Processing:**

    - Final repair costs are calculated.

    - Payments are logged in the Payments table with appropriate payment statuses (**Unpaid**, **Partial**, or **Paid**).

7. **Pickup & Closeout:**

    - The client inspects the device, clears any remaining balance, and receives their receipt.

    - The repair request status is updated to **Completed**.

8. **Reporting & Decision Support:**

- Operational data automatically populates views such as Technician_Workload, Repair_Revenue, and Most_Requested_Services for administrative oversight.

I. Introduction

The **Computer Repair Appointment and Service Management System** is a relational database solution designed to automate and streamline service shop operations. Managing repair workflows, tracking hardware/software diagnostics, monitoring technician availability, and handling billing manually often leads to lost records, unfulfilled appointments, and administrative delays.

Implementing a structured Information Management (IM) framework ensures that operational data—spanning customer profiles, device specifications, service catalogs, appointment schedules, repair statuses, and payment histories—is captured accurately, stored securely, and processed efficiently.

II. Organizational Profile and Information Environment

The repair management center provides technical support, hardware repairs, OS configuration, and malware removal services.

- **Operational Environment:** Clients bring devices in for diagnosis or pre-book repair appointments. Reception staff and service technicians collaborate to diagnose issues, execute repairs, log service charges, and issue payment receipts.

- **Information Categories:**

  - *Customer Records:* Names, phone numbers, email addresses, and home/business addresses.

  - *Device Information:* Device type (laptop/desktop), brand, model, serial number, and initial user-reported symptoms.

  - *Transaction Logs:* Appointment schedules, repair status progression, diagnostic notes, technician assignments, and final billing amounts.

    - *Catalog & Staff Data:* Service offerings, price structures, technician specializations, and daily work availability.

- **Challenges Solved:** Replacing paper logs or unorganized spreadsheets with a relational database removes record duplication, prevents unassigned repair jobs, tracks inventory demand, and provides financial visibility.

III. Information Architecture

The relational database structure organizes data across primary entities linked via Primary Keys (PK) and Foreign Keys (FK):

- **CUSTOMERS (**customer_id **PK):** Stores unique client details.
- **COMPUTERS (**computer_id **PK):** Stores hardware specs; links to CUSTOMERS (customer_id FK).
- **TECHNICIANS (**technician_id **PK):** Tracks technician contact information, specialization, and availability.
- **REPAIR_SERVICES (**service_id **PK):** Catalog of standard repair services and set prices.
- **APPOINTMENTS (**appointment_id **PK):** Manages scheduled client visits; links to CUSTOMERS and COMPUTERS.
- **REPAIR_REQUESTS (**repair_request_id **PK):** Tracks diagnoses, statuses, cost estimates, and assignments; links to COMPUTERS, TECHNICIANS, and REPAIR_SERVICES.
- **PAYMENTS (**payment_id **PK):** Tracks client payments and remaining balances; links to REPAIR_REQUESTS.

IV. Master Data Management (MDM) Framework

Master Data Management establishes single sources of truth across key business entities:

- **Core Entities Managed:** Customer Directory, Technician Profiles, and Service Catalogs.

- **Standardization Controls:** Standardizing entry formats for customer contact details and device serial numbers prevents duplicate profile creation. Centralizing service prices in REPAIR_SERVICES prevents pricing inconsistencies across work orders.

- **Redundancy Reduction:** Storing master entities separately from transactional tables (Repair_Requests, Payments) allows contact updates or fee adjustments without altering historical repair records.

V. Data Governance and Quality Policy

Data governance defines operational responsibilities and enforces data validation constraints across the database:

- **Validation Constraints:**

    - *Domain Constraints:* Restricted status fields via check constraints (chk_repair_status, chk_appointment_status, chk_payment_status).

    - *Value Guardrails:* Cost and payment fields strictly enforce non-negative values (estimated_cost >= 0, final_cost >= 0, amount_paid >= 0).

    - *Mandatory Attributes:* Critical fields such as customer names, device types, and reported problems strictly enforce NOT NULL rules.

VI. Privacy, Regulatory, and Compliance Considerations

Handling client hardware and personal contact information requires compliance with applicable privacy regulations:

- **RA 10173 (Data Privacy Act of 2012):** Explicit consent is gathered from clients during initial intake before capturing Personally Identifiable Information (PII).

- **Access Control:** Restricted database permissions ensure staff access is limited to information required for their operational roles.

- **Data Protection:** Technical safeguards prevent unauthorized extraction of customer directories or financial transaction histories.

VII. Information Security and Risk Management

Security controls protect system data against unauthorized access, hardware failures, and software threats:

- **Identified Vulnerabilities & Mitigations:**
    - *Unauthorized Data Access:* Controlled using role-based application logins and database privileges.
    - *System Failure / Corruption:* Addressed through regular automated database backup routines.
    - *Invalid Entry Errors:* Prevented using database check constraints and front-end input validation rules.

VIII. Integration and Use for Decision-Making

Integrated data flows feed into specialized database views that support operational decision-making:

- Active_Repair_Status: Tracks all uncompleted repairs to prevent backlog delays.
- Technician_Workload: Aggregates active task counts per technician to ensure balanced work distribution.
- Repair_Revenue: Computes total paid amounts versus remaining balances per repair order.
- Most_Requested_Services: Tracks service frequencies to inform parts inventory planning.

IX. Summary of Findings and Recommendations

- **Findings:** Implementing a centralized relational structure effectively eliminates lost repair jobs, balances technician assignments, and automates tracking of client payment balances.

- **Recommendations:**

1. Maintain strict front-end data validation matching database check constraints.

2. Perform periodic audits on active technician workloads to optimize turnaround times.

3. Ensure routine offsite backups of system databases to guarantee disaster recovery capability.

Appendices

Appendix A: RACI Matrix (Data Governance)

Data Domain / Process	**Shop Manager**	**Front Desk Staff**	**Technicians**	**System Admin**
Customer Records Management	Accountable (A)	Responsible (R)	Consulted (C)	Informed (I)
Repair Status Updates	Informed (I)	Responsible (R)	Responsible (R)	Accountable (A)
Service Pricing & Catalog	Accountable (A)	Informed (I)	Consulted (C)	Responsible (R)
Payment & Billing Logs	Accountable (A)	Responsible (R)	Informed (I)	Support (S)

Appendix B: Risk Matrix (Information Security & Risk Management)

Threat / Risk Event	**Likelihood**	**Impact**	**Risk Level**	**Mitigation Strategy**
Unauthorized Data Access	Medium	High	High	Enforce role-based access control and strict user authentication.

Threat / Risk Event	Likelihood	Impact	Risk Level	Mitigation Strategy
Data Loss via Hardware Crash	Low	Critical	High	Implement automated daily database backups.
Malware / System Infection	Medium	High	High	Maintain workstation antivirus software and restrict external media usage.
Invalid Data Entry (Negative Costs)	High	Low	Medium	Apply database check constraints (CHECK >= 0).

Appendix C: Integration Matrix (Integration & Decision-Making)

Operational Workflow	Integrated Modules	Data Output / Report Generated	Decision Support Value
Appointment → Repair Request	Customers, Computers, Appointments, Technicians	Active_Repair_Status, Appointment_Schedule	Ensures timely technician assignment and prevents double-booking.
Diagnosis → Repair Execution	Computers, Technicians, Repair_Services, Repair_Requests	Pending_Repairs, Technician_Workload	Helps supervisors distribute active repair tasks evenly.
Repair Completion → Payment	Repair_Requests, Payments, Repair_Services	Repair_Revenue, Most_Requested_Services	Tracks real-time revenues, outstanding balances, and inventory demand.